

Technical Proposal for the
ENSIS4020AJZ_SERIES
FIBER Laser System

Prepared for
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Att: Darrell Thompson,

Technical Proposal Number: 160-95856
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*The ENSIS4020AJZ Fiber Laser is one of the original
engineered solutions dedicated to precision sheet metal
fabrication.*

Visit our website at www.amada.com

DISCLAIMER:

This proposal/quotation is intended to outline the cost and specifications for the equipment outlined herein. In no case beyond the twelve months parts and labor, as outlined in Amada's warranty for new equipment, shall anything in this proposal/quotation be construed as a warranty.

In no event shall Buyer or Seller be liable to the other party for anticipated profits or for incidental or consequential damages resulting from an equipment sale. Buyer or Seller's liability on any claim of any kind for any loss or damage shall in no case exceed the price allocable to the articles or unit, which gives rise to the claim. Any action resulting from any breach on the part of Seller as to the articles delivered under a Purchase Order must be commenced within one (1) year after the cause of action has accrued.

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1. Introduction

1.1 THIN TO THICK FIBER

Since 1980, Amada has been providing industrial laser cutting machines to the fabricating market. As the world leader for the fabricating machine tool industry, Amada understands the need for designing equipment to meet the ever changing demands on manufacturing, while never compromising the quality and dependability that have kept Amada number one over the years.

The ENSIS4020AJZ Series is a revolutionary advancement in fiber laser cutting technology. These high speed fiber lasers use on average less than 1/2 of the electricity of a 4000 watt CO2 laser yet cut up to 4-5 times faster in thin materials and matches speed and quality in thick materials up to 1 inch! The ENSIS is the first fiber laser designed to handle the full range of mild steel, previously attainable only with 4kW CO2 machines.

Amada uses our own fiber technology, developed with Lumentum. This has resulted in better quality laser diodes with the highest output power and efficiency of any laser available today.

By developing our own fiber laser, Amada has been able to create the proprietary features, which made the ENSIS a reality. The ENSIS is the ultimate laser cutting system.

Key Features of the ENSIS:

- Continuously variable beam quality for full range of cutting capability
- Amada designed fiber technology
- Lowest cost operating fiber system
- Highest efficiency fiber laser system
- Nozzle changer, WACS™, Pierce Monitor, SPF Pierce Control



For illustrative purposes only, some safety features and guards may have been modified or removed from the Amada equipment pictured in this quotation. You should not rely on these pictures for specific machine applications. Instead, contact your



representative at Amada America, Inc. regarding machine applications and operating specifications.

All machine tools require proper safety equipment, guarding and operation based on ANSI, OSHA and other applicable Federal, State and Local rules, regulations and requirements.

It is the responsibility of the owner/operator to ensure that suitable safety measures are implemented and maintained.

2. Machine Description

2.1 TURN KEY PACKAGE

All of Amada laser-cutting systems are turnkey packages. The package includes the machine motion system, chiller, dust collector, gas regulators and connection kit. The end user need only supply consumables including electrical and air connections, water, laser and cutting assist gases. Amada recommends all machines be placed on a level, continuous slab. Consult with the Amada laser consultant for questions regarding the foundation requirements.

All components are serviced and supported by Amada's highly trained personnel. Amada's off line programming software and hardware can be supplied with the machine as an option.

2.2 MOTION SYSTEM

The cutting head and drive mechanisms are mounted on the exceptionally rigid frame design. The fiber oscillator is also mounted directly to the frame further simplifying the install process and allowing for a small floor space requirement.

The material remains stationary on the cutting table as the cutting head moves in the X, Y and Z-axis. This high precision three axis drive system consists of re-circulating ball lead screw in the Z axis with dual high speed, helical rack and pinion drives in the X & Y-axis. The helical R & P system offers increased machine acceleration by allowing for more torque to be used. The result is faster part processing with a quieter and smoother drive system. In conjunction with LM guides the motion accuracy and positioning speeds provides exceptional performance. The Z-axis motion is CNC controlled. The standard high-speed capacitance cutting head maintains a constant nozzle gap regardless of the contours of the work piece.



2.3 THE AMADA ENGINEERED FIBER OSCILLATOR

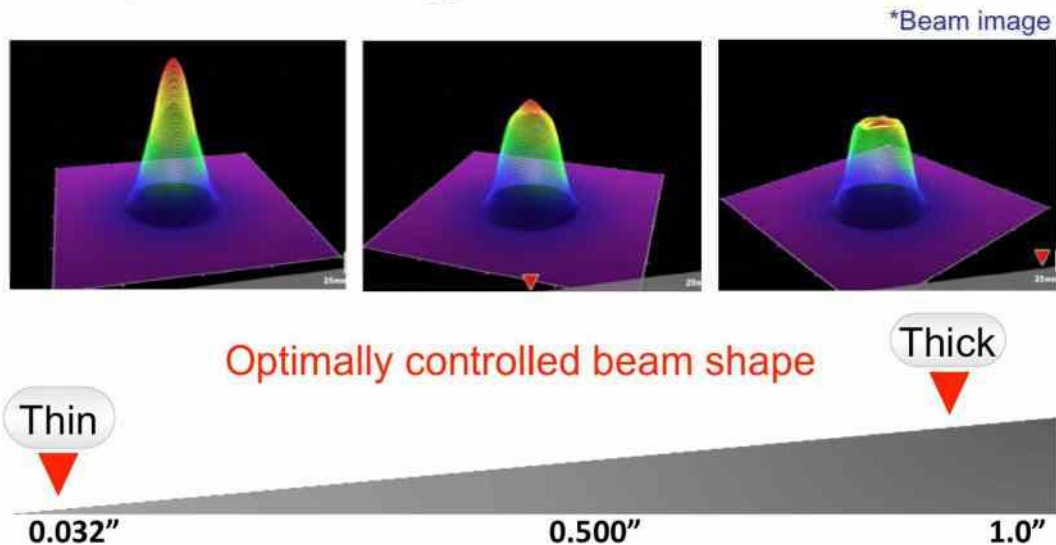
The AJ SERIES on the ENSIS is Amada's 3rd generation fiber engine, utilizing fewer modules than previous versions. This creates a simpler design with better energy efficiency and increased reliability. The specially designed diodes boast the highest power output along with the highest electrical efficiency of any laser diode today.

The AJ SERIES is a true fiber laser source. This simple design has many benefits:

- No laser gas
- No mirrors to align, clean, or replace
- 1/2 the electrical consumption & 3 times the energy efficiency of comparable CO₂
- Better over all efficiency than any other method of beam generation

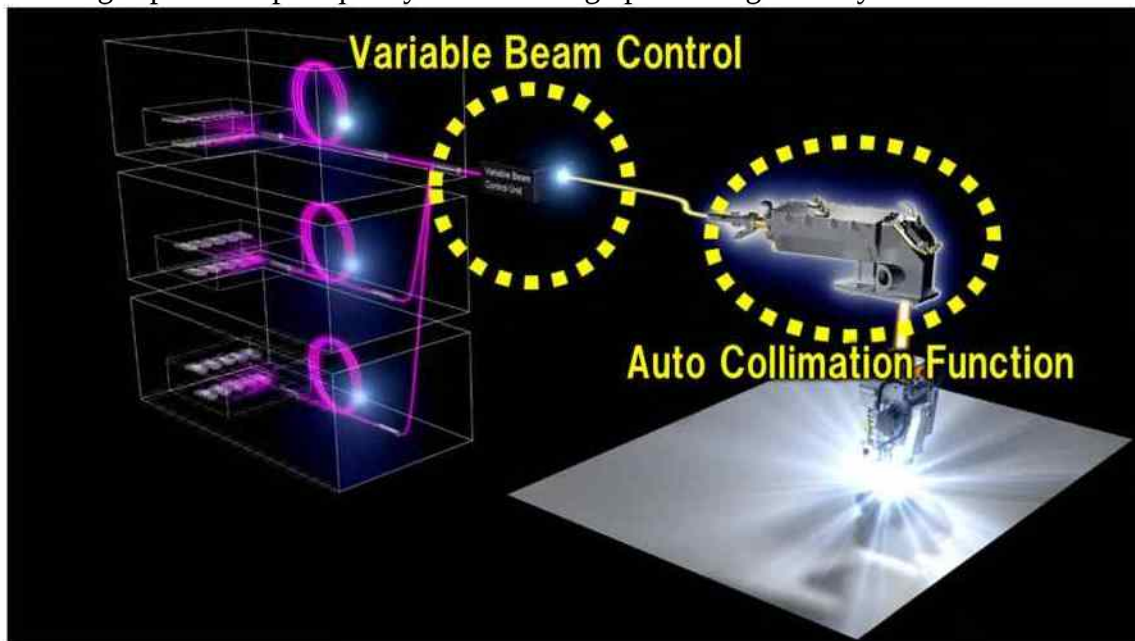
2.4 THE ENSIS TECHNOLOGY

ENSIS is Latin for “sword”. This patented technology allows for automatic adjustment of the beam quality to take full advantage of the fiber laser’s low cost of operation and high speeds in thin material, and also excellent edge quality and speeds of thick plate, previously only achievable by CO2 lasers. The mode is changed automatically to optimize the cut quality throughout the entire range of the materials being processed.



2.5 AUTO COLLIMATION FUNCTION

The Auto Collimation Function allows the machine to seamlessly optimize the beam diameter and focal point based on material and thickness. This coupled with the Variable Beam Control unit enables the Ensis to cut thin to thick materials with high speed and part quality as well as high processing stability.



2.6 FLYING “FIBER” BEAM DELIVERY SYSTEM

The fiber laser can cut faster than any other method in thin material. The flying optic or “fiber” design is the best motion system to take advantage of these cutting speeds. Why? Since the material remains stationary, the motion system is always moving a known mass, the gantry. This allows for precise calculations in the acceleration and deceleration potential on all materials, which takes full advantage of the high-speed cutting potential of the fiber laser.

The flying fiber beam delivery system is a very simplified design, which takes the laser beam from the oscillator to the Z-axis via a single fiber cable called the “process fiber”.

2.7 FUME AND PARTICULATE EXTRACTION

A CNC controlled damper and plenum system under the cutting area provides superior dust and smoke extraction. This system assures consistent, high-speed cutting and excellent part quality. The system is designed to operate in zones, so the dampers are open directly beneath the area where the cutting head is, while the dampers elsewhere remain closed. The enclosure further helps to keep dust and particulate in the cutting area for more efficient removal.

2.8 AMNC-3i CONTROL

This new, large format control has a 21.5” screen size with 16:9 aspect ratio and a 1920 X 1080 HD resolution

A simple, intuitive touch screen makes navigating the ENSIS-AJ easier than ever. The entire cutting library for the laser is stored in the controller’s memory. The operator interface panel is mounted to the front face of the enclosure and pivots for better viewing by the operator. This design allows for a cleaner installation with a more compact footprint.

The AMNC control's touch screen and color graphics monitor can display all of the laser and machine operating conditions. Visual thumbnail views of nests stored in the network make selecting a program easier than ever and larger display of the nest makes easier editing on the shop floor if desired. Operating conditions include but are not limited to: cutting data library, axis position, feed rate, program execution, programs in memory, machine alarms, machine parameters, diagnostics, and maintenance reminders.



2.9 PIERCING

For thin material, the laser pierces the material almost instantaneously providing turret-like throughput. For thicker materials, canned cycles vary the piercing pulse frequency and duty cycle progressively by pre-determined steps providing high material removal. Using these read-only, memory-canned cycles reduces the piercing time significantly and they can be edited for specific program parts.

2.10 CUTTING DATA LIBRARY

The laser system features an integrated cutting data library with parameters for many common materials. The AMNC control can hold up to 1,000 different material-cut conditions. The machine operator can make modifications to the library at the control. The library management software can list all of the available material files or can be requested to display only those files that meet a certain criteria.

Each of the 1,000 possible material files is identified with a descriptive name. Each file includes up to 10 different cutting parameter sets of data that may be assigned to different applications (precision, standard, high speed, etching, etc).

The operator can override all of the parameters in the material data file "on-the-fly". Parameters can be changed permanently or temporarily. This allows the operator to quickly compensate for a difficult batch of material without producing scrap parts.

Head movement (Fast, Medium, Slow), and piercing functions can also be controlled at the controller in the case productivity can be increased or the potential for a head crash exists.

2.11 ADVANCED CUTTING TECHNOLOGIES

Amada has pioneered several of the industries most important laser cutting techniques. WACS™ provides a stream of water mist to keep the material cool and produce accurate parts in thick steel. CleanCut™ provides a high quality, oxide-free, weld and paint ready edge for mild steel, stainless steel, aluminum, and titanium. CleanCut™ utilizes a high-pressure inert assist gas, typically nitrogen to prevent oxidation of the cut edge. AirCut™ uses clean dry and filtered air to process mild steel, aluminum and stainless steel if the application permits. Using AirCut™ can increase cutting speeds by as much as 3 times faster over standard cutting with oxygen.

2.12 AUTOMATIC NOZZLE CHANGER

The ENSIS utilizes an 8 station nozzle changer that automatically changes the nozzle based on the cut condition being used in the control. This eliminates any downtime associated with an absent operator causing the machine to remain idle. The auto-nozzle change feature dramatically increases Green Light On (GLO) time.

16 station nozzle changer is standard on 12kw Ensisi machines

2.13 NOZZLE CLEANER & CALIBRATION PLATE

The automatic nozzle cleaner will clean the nozzle on the cutting head either on command from the operator or as part of a set cycle time. Factors such as pierce count, hours of cut time, or machine on time can be set to initiate a nozzle cleaning. The calibration plate is in place to automatically calibrate the head after a nozzle change.

2.14 AUTOMATIC CORNER CONTROL

Automatic corner control selects appropriate laser power level, pulse frequency and pulse duty cycles to provide sharp burn free corners. Different parameters can be selected, according to the angle of the corner to be processed.

2.15 HIGH SPEED SHUTTLE TABLE

The high-speed shuttle table allows for external loading, unloading and set-up of material. Each table incorporates manual clamps so that different materials of different thickness can be clamped to the same table. This prevents the material from sliding over the table support grid when the shuttle tables are exchanged.

Micro tabbing of parts can be greatly reduced and even eliminated in many cases.



2.16 SLAT TABLE

The machine features a consumable support slats that can be modified to suit a particular application. Tabbing may still be required for certain parts based on nest direction. The slat sections can be removed for processing large, thicker parts to reduce the rate that the grid is consumed.

2.17 EASY SHEET LOAD AND UNLOAD

The height of the table is designed so that the operator can load or unload material with safety and comfort. For automated production needs, the machine can be automated with a fork load unload system or a multiple shelf pallet exchange system for maximum machine utilization.

2.18 AUTOMATIC RETRY FUNCTION

If the cutting head detects an obstacle preventing the head from cutting, an automatic retry function is initiated. The CNC retracts the cutting head, retraces the cutting path prior to the obstacle and then re-processes the problem section. This cycle can be repeated a specified number of times (usually three) to try and recover the cut. If the retry function fails to complete the cut, the machine can be programmed to skip to the next part to be processed. This feature is particularly useful for processing parts with certain complicated geometries, materials that are difficult to process or lesser quality materials.

2.19 HIGH-SPEED CAPACITIVE CUTTING HEAD SYSTEM

The machine features an all-in-one, high-speed, non-contact, capacitive cutting head suitable for a variety of cutting applications. The head can follow distorted material. A built-in macro enabled by the operator automatically calibrates the head to ensure consistent cutting. The focus lens can be quickly removed and replaced in the cutting head via the simple cartridge design and does not require any tools.

The focus lens assembly is mounted in a B axis drive mechanism that allows for CNC adjustment. Adapters are provided to allow the use of 190mm and 190AXmm lenses. The cutting head assembly delivers the assist gas to the work piece co-axial to the laser beam. The cutting head assembly incorporates a breakaway collar to minimize head damage in the case of a collision.

2.20 CHILLER

The chiller system is a refrigerated water to air, heat exchanger that is used to keep the laser resonator, and optics at a constant temperature. The chillers used by Amada are designed to consume less energy than conventional chillers. Customer supplied distilled water is required to be added to the chiller at initial installation and for routine maintenance top-off.

Hoses and fittings are provided to connect the chiller to the laser machine.

2.21 DUST COLLECTOR

A dust collector is provided to collect the fumes and air borne particulate matter created by the cutting process. A dust collector is essential for effective and safe operation of a laser-cutting machine.

2.22 ENCLOSURE

The machine is supplied with an enclosure around the cutting table area. An enclosure around the cutting area helps improve dust collection and protects from scattered radiation. The cutting area is still accessible through the interlocked doors and the cutting process can be viewed through the windows around the machine.

2.23 INTERFACE DEVICES

The AMNC-3i incorporates numerous interface devices. All of the following are provided as standard items.

LAN: 2 ports 1GBPS

USB: 2 ports @ 3.0 – 6 ports @ 2.0

Serial: 2 ports

Expansion Slots: 2 slots PCI-Express

2.24 GAS REGULATOR AND CONNECTION KIT

The machine is supplied with regulators and hoses to connect the machine to standard assist gas bottles. Regulators and hoses are supplied to connect the machine to the separate assist gases required to process material. To maintain high nitrogen assist gas pressures, customer supplied additional fittings and components may be required.

2.25 CONSUMABLES

The machine is supplied with focus lenses and cutting nozzles and slat tables. These items will need to be cleaned and replaced at a frequency dependent on the application and processing techniques. Additional consumables such as oils, lubricants and other items will need to be purchased by the customer on an as needed basis.

3. System Specifications

3.1 STANDARD SPECIFICATIONS:

FEATURES	6kW	9kW	12kW
ENSIS Vario Technology	✓	✓	✓
Auto Collimation Function	✓	✓	✓
Flying "Fiber" Beam Delivery System	✓	✓	✓
AMNC-3i Control	✓	✓	✓
Cutting Data Library	✓	✓	✓
Advanced Cutting Technologies	✓	✓	✓
8 Sta. Automatic Nozzle Changer	✓	✓	○
16 Sta. Automatic Nozzle Changer	○	○	✓
Nozzle Cleaner & Calibration Plate	✓	✓	✓
Automatic Corner Control	✓	✓	✓
High Speed Shuttle Table	✓	✓	✓
X-Axis Slug Conveyor	○	○	✓
Automatic Retry Function	✓	✓	✓
High-Speed Capacitive Cutting Head System	✓	✓	✓
Chiller	✓	✓	✓
Dust Collector	✓	✓	✓
Class 1 Enclosure	✓	✓	✓
Interface Devices	✓	✓	✓
Gas Regulator and Connection Kit	✓	✓	✓
OVS (Optical Verification System)	○	○	○
KEY:	✓	= Included	
	○	= Factory Option	



3.2 SUPPLIED COMPONENTS

The ENSIS4020AJ is supplied with all necessary components. Detailed specifications are outlined below.

Motion Package	ENSIS4020AJ – 160" x 80" Table
Laser Resonator	Amada AJ SERIES with beam variation and ACF
CNC Controller	AMNC-F
Chiller Cooler	Orion or equivalent
Dust Collector	Camfil or suitable equivalent.
Gas Kit	Laser and assist gas connection kit.

3.3 MOTION PACKAGE

Travel Method	Stationary table, X, Y and Z-axis movement for cutting head
Control Method	X, Y and Z simultaneous 3 axis control
Drive Motors	Fanuc AC servo motors
Maximum Sheet Size	80" (Y) x 160" (X)
Maximum Axis Travel	80.70" (Y) x 160.23" (X) x 3.93" (Z)
Maximum Axis Positioning Speed (X, Y)	4,724 inches per minute per axis (6,693 inches per minute 45° vector.)
Maximum Speed Z-Axis	3,149 inches per minute
Repeatability	±0.0002"
Maximum Material Weight	3,000 lbs.
Assist Gas	4 ports with NC pressure Control
Electrical Requirements	200V, 3 Phase, 60Hz

3.4 CNC CONTROLLER

Model	AMNC-3i
Display	TFT Color LCD
Control Function	X, Y, Z-Axis Simultaneous and Focus Lens B-axis, and Laser Power control (CW, Gated Pulse, Pierce Routines etc.).
Data Input Method	Manual (MDI), Edit, LAN, USB, Serial, Expansion
Minimum Command Unit	0.0001" (X, Y, Z)
Minimum Travel Unit	0.0001" (X, Y, Z)
Standard Memory	2 mb
Compact Flash Drive	2 gb
Operating Modes	Automatic and Manual
Electrical requirements	200V 3 Phase, 60Hz

3.5 LASER RESONATOR

Model	Amada AJ SERIES with beam variation control
Resonator Style	LD excited Yb fiber
CW Output Power	6000W, 9000W, 12000W
Peak Pulse Output Power	6000W, 9000W, 12000W
Power Stability	# 2%
Laser Wave Length	1.08 μ m
Laser Power Mode Selection	CW, Gated Pulse (CNC Controlled)
Laser Beam Mode Output	Multimode Mode
Beam Divergence	< 3mm-mrad
Pulse Frequency	1 – 10000Hz
Pulse Duty	0 – 100%
Chiller Water Requirement	6kw > 100 liters per min.
	9kw > 130 liters per min.
	12kw > 152 liters per min.
Interlocks	Electrical, Mechanical, and Chiller
Electrical Requirements	6kw - 200 VAC, 3 Phase, 60Hz 28.1kVA max.
	9kw - 200 VAC, 3 Phase, 60Hz 35kVA max.
	12kw - 200 VAC, 3 Phase, 60Hz 43.4kVA max.

3.6 CHILLER

6kW Model	Orion RKE7502-VA-DIBP
Chiller Cooling Capacity	5.91 tons (70,972 BTU/hr)
Electrical Requirements	200 VAC, 3 Phase, 60Hz, 25A, 11.4 kVa

9kW Model	Orion RKE110002-VA-DIBP
Chiller Cooling Capacity	8.24 tons (98,952 BTU/hr)
Electrical Requirements	200 VAC, 3 Phase, 60Hz, 25A, 19.5 kVa

12kW Model	Orion RKE150002-VA-DIBP
Chiller Cooling Capacity	11.37 tons (136,486 BTU/hr)
Electrical Requirements	200 VAC, 3 Phase, 60Hz, 25A, 22.8 kVa

3.6 DUST COLLECTOR

Model	Camfil or Suitable Equivalent
Discharge Port Size	11.8" Diameter
Power Requirements	208 VAC, 3 Phase, 60Hz, 12 kVA

4. Description of Machine Options

4.1 OVS (OPTICAL VERIFICATION SYSTEM)

The OVS is an ideal feature for those manufacturers' still prepunching sheets of material, but who still want the perimeter of the parts cut by the laser. The OVS utilizes a built-in camera at the cutting head that measures the center area of two holes. It then calibrates the laser program to match the punch program within $\pm 0.002"$. (Factory Installed Option Only)

4.2 STEP DOWN TRANSFORMER

The machine is configured for 200V 3-phase operation. If your shop has 230V or 460V incoming voltage, a suitable step down transformer is required. A separate transformer can be purchased for your voltage requirement. This can be purchased from your local vendor or purchased as a separate item from Amada.

4.3 ASSIST GAS GENERATORS

Amada offers a full line of N2 assist gas generation systems from the low volume, lower pressure membrane designs, all the way to high demand, high purity and high pressure PSA systems. Systems will be designed and quoted based on application requirements.

4.4 SOFTWARE

Amada's complete solution involves the excellent software options for this and all Amada machines. The Dr. ABE brand of software from Amada takes the guesswork out of programming parts that need to be blanked and bent. Part accuracy is verified before it ever goes to the floor. All software programs are designed for ease of use and are capable of generating NC code output for all Amada machines using the latest laser technology. As new machines enter the market place software drivers are developed to meet fabricators needs improving productivity and quality.

vFactory is the ultimate software for knowing exactly how much utilization is being achieved on your Amada equipment. It monitors all factors, including setup time, alarms, planning, and green light on time. From this you can get weekly, monthly and yearly reports for a proper picture of your shops utilization. Then you can make adjustments to further increase your machines green light on time.

5. After Sales Service

5.1 SERVICE AND WARRANTY

Prices quoted include twelve (12) months parts and labor warranty to the machine, laser, CNC and other supplied components, due to defects in material or workmanship. All supplied component parts are warranted against defects for a period of twelve (12) months and will be repaired or replaced at Amada's option during that period.

5.2 INSTALLATION

Your Amada laser consultant will review the laser pre-installation guide with you.

Assuming all site requirements are met as per Amada's pre-installation requirements, an Amada service engineer will be scheduled for installation. The engineer will take the time necessary to make sure that all the necessary connections are made to bring the machine on-line and confirm that the machine is operating to specification.

Operators will be trained in basic operation and the service engineer will make routine maintenance procedures training. The service engineer accomplishes this once the machine is operational.

*Please note that installation cannot and will not take place until all pre-installation requirements are met.

5.3 CUSTOMER SUPPORT

Our engineers, with years of laser processing experience are an invaluable asset to help you progress up the learning curve as rapidly as possible. Process application and service engineers are located on the East Coast and the West Coast to respond to your telephone inquiries. Should your particular application be such that a solution cannot be found over the phone, a service engineer is promptly dispatched to your facility.

5.4 TRAINING

Training Overview:

Training for one operator is included with the purchase of every Amada laser machine. Primary training is conducted at an Amada facility in Schaumburg, IL or Buena Park, CA depending on the available class schedule and machines. Follow-up training is conducted on-site at the customer facility. Primary training is structured in two phases. Phase one focuses on safety, basic machine operation, maintenance, understanding of basic machine code, and laser cutting basics for thinner materials. Phase two focuses on cutting operation, machine setups, options and other applications needed to operate the laser under normal, everyday conditions, emphasis is on thicker materials. This phase will also be geared toward specific customer cutting needs. Follow-up training involves an Amada instructor or application engineer who will visit the customer at a scheduled date, typically around 4 weeks after the phase two portion of the training has been completed. This follow-up phase is to answer any newly developed questions and to assist in more direct customer applications.

Time Frame of Training:

NOTE: It is Amada's experience and strong recommendation that Phase One and Two training be conducted at a specified Amada facility. This has proven to offer the trainee the best opportunity to learn the new equipment with the least amount of distraction. Also, training at an Amada facility will be conducted on the same or comparable model laser machine based on availability in the selected Amada location. In the case Amada does not have the same or comparable model in one of the select locations; Amada will conduct on-site training at the customer facility at no additional charge (training class capacity to be limited to those that completed Phase One).

Training time for Phase One and Phase Two is for three and 1/2 days typically starting on a Tuesday morning and ending on a Friday afternoon, unless other arrangements are made. Training for phase one & two may be done on-site at the customer location for an additional charge. On-site training schedule is subject to trainer availability. The follow-up phase is generally used to cover specific customer application questions with the time frame based on scheduling, two days being the standard time frame. Specific arrangements for other time frames can be purchased. Training for this period is to be conducted on-site at the customer facility using customer machine and materials. Additional on-site follow-up training is available at an additional charge.

Number of People to be Trained Included in Machine Sales Price:

In order to insure the quality of training conducted for everyone in an Amada facility, Amada limits the class size per session to no more than 5 students to allow for hands on experience and comprehension. Due to the volume of laser sales it is difficult to accommodate more than one student per company. Additional students from one company may be permitted at an additional charge. The addition of more the one student is subject class size availability.

If it is necessary to train more than two operators from one facility Amada recommends purchasing on-site training for your facility. On-site training from Amada will accommodate up to five personnel from your company to be trained at your facility once the laser is installed and operational.

Pricing Information

Amada, ENSIS4020AJZ 6kw FIBER Laser System

F.O.B.* Shipping Points \$849,000.00

Amada, ENSIS4020AJZ 9kW FIBER Laser System

F.O.B.* Shipping Points \$999,000.00

Amada, ENSIS4020AJZ 12kW FIBER Laser System

F.O.B.* Shipping Points \$1,135,000.00

* All Prices quoted are FOB Shipping Point. Supplemental equipment and accessories included in this quotation originate in Amada and/or vendor warehouses. All supplemental equipment and accessories are also sold FOB * shipping point. Such items include, but are not limited to chillers, dust collectors, transformers, safety mats, additional conveyors, installation kits, optics, software and drivers.

Optional Equipment Pricing

Automation TBD

OVS (Optical Verification System) factory install only \$30,000

Amada Software TBD

Amada N2 Generation Systems TBD

Amada Gas Mixer TBD

Step Down Transformer TBD

On-site Training - 3.5 days \$6,000

Additional Student at Amada School - 3.5 days \$1,000

1/8" Copper Slat Option TBD

1/8" Slat Copper Hybrid TBD

Slat Cleaner PN 97030070 \$6,800

Ball Transfer Option (not available with automation) TBD

7. Terms, Delivery and Financing

7.1. Delivery

If possible, a firm delivery date will be established at the time the purchase order is issued to Amada. If a firm delivery date cannot be established the order is subject to the terms and Conditions in the next section. If the purchaser is unable to take delivery within 2 weeks after the agreed upon delivery date, the machine will be re-routed, and a new delivery date will be established.

7.2. Terms and Conditions

Prices stated in this quotation are valid for (30) days.

Prices do not include any State, Federal or local taxes which are now in effect or may be imposed.

Payment terms in the USA are subject to the following conditions:

- **Standard terms:** Standard specification equipment with confirmed ETA
 1. 10% Deposit with PO.
 2. 80% Due upon Delivery (not applicable with ACC financed terms).
 3. 10% net or ACC finance terms 30 days after installation sign-off.
- **ETA TBD terms:** Standard specification equipment with unconfirmed ETA.
 1. \$5,000 Deposit with PO.
 2. Balance of 10% deposit due with confirmed ETA.
 3. 80% due upon delivery (not applicable with ACC financed terms).
 4. 10% net or ACC finance terms 30 days after installation sign-off.
- **Special order terms:** Non-standard specification equipment.
 1. 30% Non-refundable deposit required with PO.
 - Note: Special order machines cannot be reserved without deposit.
 2. 60% due upon delivery (not applicable with ACC Financed terms).
 3. 10% net or ACC finance terms 30 days after installation sign-off.

A 1.5% interest, per month will be assessed on all balances not paid within the 30-day period.

All shipping charges are billed freight collect, F.O.B. Shipping Point

International Shipments require advance payment or Letter of Credit

7.3. Financing

Low cost financing is available to those that qualify in the USA. Further information is available upon request.

We thank you for the opportunity to offer this technical proposal and look forward to serving you in the future.

Sincerely yours,

AMADA AMERICA, INC.

Michael G. Decono, Sales Engineer

ENSIS4020AJZ

Prepared exclusively for:

**LOS ALAMOS NATIONAL LAB
Darrell Thompson,**